

1. What elemental symbol represents the alkali metal in the fourth period of the periodic table?
 - A) K
 - B) Sr
 - C) Br
 - D) Rb
 - E) Ca
2. Three of your five professors in a given semester talk about their pets a lot during class. All three pet-loving professors teach science courses, and the remaining two professors teach English and history, respectively. You hypothesize that science teachers like pets more than professors in other areas and decide to use the scientific method to test your hypothesis. What should you do next?
 - A) Come up with a theory to explain why teaching science is related to loving pets.
 - B) Design an experiment to test your hypothesis.
 - C) Propose several alternative hypotheses.
 - D) Refine your hypothesis.
 - E) Publish your findings.
3. There are three naturally occurring isotopes of silicon, and the atomic mass and percent abundance of two are listed below. What is the atomic mass of the third isotope (in atomic mass units) if the atomic weight for silicon is 28.08538 amu?

^{28}Si	92.23% abundant	27.97693 amu
^{30}Si	3.09% abundant	29.97277 amu

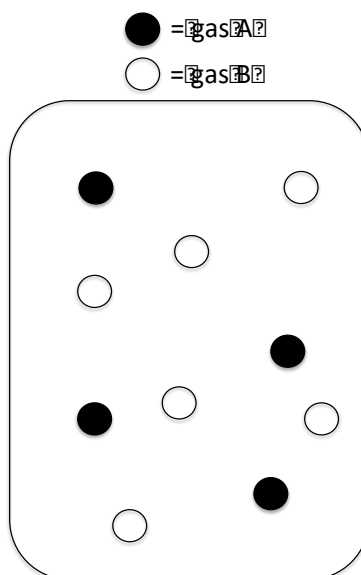
 - A) 25.80312 amu
 - B) 28.0900 amu
 - C) 28.97649 amu
 - D) 29.07524 amu
 - E) 29.99584 amu
4. What is the chemical symbol for an atom with 79 protons, 118 neutrons, and 76 electrons?
 - A) $^{118}\text{Os}^{3+}$
 - B) $^{79}\text{Os}^{3-}$
 - C) ^{118}Au
 - D) $^{197}\text{Au}^{3+}$
 - E) $^{197}\text{X}^{3+}$

5. What is the mass (in grams) of 8.5×10^{24} molecules of water?
- A) 0.0039 g
 - B) 2.8 g
 - C) 18 g
 - D) 250 g
 - E) 760 g
6. The detonation of liquid nitroglycerin, $\text{C}_3\text{H}_5\text{N}_3\text{O}_9$, produces four gaseous products as shown below. What is the coefficient for CO_2 when the equation is balanced using the smallest whole number coefficients?
- $$\text{C}_3\text{H}_5\text{N}_3\text{O}_9(\text{l}) \rightarrow \text{CO}_2(\text{g}) + \text{N}_2(\text{g}) + \text{O}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$$
- A) 3
 - B) 4
 - C) 9
 - D) 12
 - E) 22
7. Which statement is true regarding ionic and covalent bonding?
- A) Covalent bonding is based on the transfer of electrons from a nonmetal to a metal.
 - B) The compound NH_3NO_3 is covalent because it contains only nonmetals
 - C) Ionic bonding is based on sharing of electrons between two nonmetals.
 - D) Covalent bonding is based on sharing of electrons between two metals.
 - E) Ionic bonding is based principally on the attraction of a cation to an anion.
8. What is the name of the compound $\text{Mn}(\text{Cr}_2\text{O}_7)_2$?
- A) Manganese(IV) dichromate
 - B) Magnesium dichromate
 - C) Manganese(II) dichromate
 - D) Manganese(II) dichromiumheptoxide
 - E) Manganese(II) dichromide

9. Complete combustion of a compound containing only carbon and hydrogen produced 33.01 g of CO_2 and 13.51 g of H_2O . The mass of 2.50 moles of the hydrocarbon is 140.3 g. What is the molecular formula of the hydrocarbon?
- A) C_4H_8
B) CH_2
C) CH
D) C_9H_9
E) $\text{C}_{10}\text{H}_{20}$
10. The reaction of Fe_2O_3 with CO produces $\text{Fe}(\text{s})$ and $\text{CO}_2(\text{g})$. What mass of Fe_2O_3 is required to produce 5.00 kg of $\text{Fe}(\text{s})$ if the percent yield for the reaction is 78.5%? You may assume that $\text{CO}(\text{g})$ is present in excess.
Molar masses (in g/mol): $\text{Fe}_2\text{O}_3 = 159.70$, $\text{CO} = 28.01$, $\text{Fe} = 55.85$, $\text{CO}_2 = 44.01$
$$\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \rightarrow 2\text{Fe}(\text{s}) + 3\text{CO}_2(\text{g})$$
- A) 7.15 kg
B) 8.69 kg
C) 9.11 kg
D) 14.3 kg
E) 18.2 kg
11. In a certain experiment, 8.0 moles of NH_3 react with 8.0 moles of O_2 . How many moles of $\text{NO}(\text{g})$ are formed, and how many moles of the excess reactant remain?
- $$4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$$
- A) 8.0 mol NO produced, there is no excess reagent
B) 6.4 mol NO produced, 6.4 mol NH_3 remain
C) 8.0 mol NO produced, 6.4 mol NH_3 remain
D) 6.4 mol NO produced, 1.6 mol NH_3 remain
E) 14.4 mol NO produced, there is no excess reagent
12. Which statement is true regarding sodium phosphate?
- A) It is a strong electrolyte that dissociates but does not dissolve in water.
B) It is a nonelectrolyte that neither dissolves nor dissociates in water.
C) It is a strong electrolyte that both dissociates and dissolves in water.
D) It is a weak electrolyte that dissolves but does not dissociate in water.
E) It is a strong electrolyte that dissolves but does not dissociate in water.

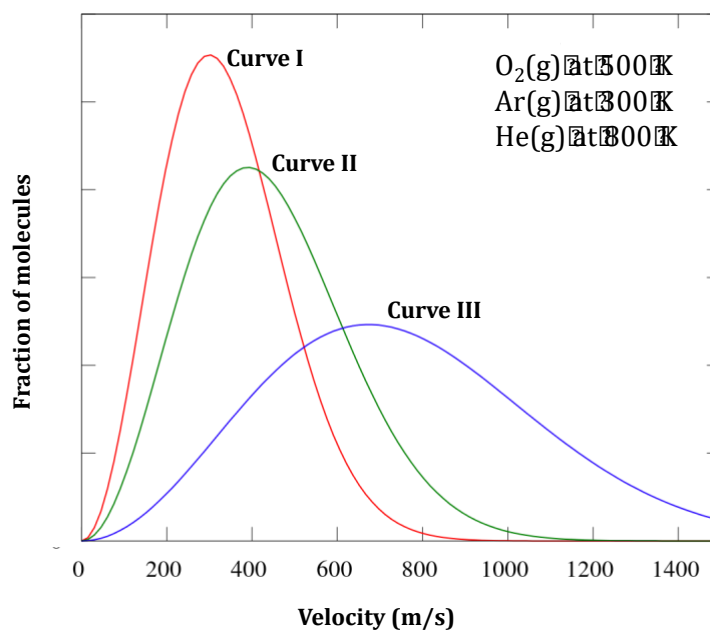
13. When two aqueous solutions of ionic compounds are mixed, a precipitate may form. Alternatively, it is possible that no reaction occurs. For which pair of compounds does a reaction not occur?
- A) $\text{Pb}(\text{NO}_3)_2(\text{aq})$ and $\text{KCl}(\text{aq})$
 - B) $\text{BaCl}_2(\text{aq})$ and $\text{Na}_2\text{SO}_4(\text{aq})$
 - C) $\text{LiOH}(\text{aq})$ and $\text{Fe}(\text{ClO}_4)_3$
 - D) $\text{KI}(\text{aq})$ and $\text{Sr}(\text{NO}_3)_2(\text{aq})$
 - E) $(\text{NH}_4)_3\text{PO}_4(\text{aq})$ and $\text{Ca}(\text{OH})_2(\text{aq})$
14. What volume (in mL) of 1.36 M Na_2SO_4 must be diluted with water to prepare 275 mL of a solution that is 0.0350 M in Na^+ ions.
- A) 0.0173 mL
 - B) 3.54 mL
 - C) 7.08 mL
 - D) 10.7 mL
 - E) 93.6 mL
15. A sample of SO_2 gas occupies 675 mL at STP. What is the pressure, in mm Hg, of the gas if the volume is increased to 825 mL at 25.0 °C?
- A) 0.622 mm Hg
 - B) 0.893 mm Hg
 - C) 201 mm Hg
 - D) 225 mm Hg
 - E) 679 mm Hg
16. Sodium azide, NaN_3 , is used to produce $\text{N}_2(\text{g})$ to inflate vehicle airbags. What mass of NaN_3 is required to produce 37.0 L of $\text{N}_2(\text{g})$ at 1.00 atm and 295 K?
Molar masses (in g/mol): $\text{NaN}_3 = 65.02$, $\text{N}_2 = 28.02$, $\text{Na} = 22.99$
- $$2\text{NaN}_3(\text{s}) \rightarrow 3\text{N}_2(\text{g}) + 2\text{Na}(\text{s})$$
- A) 1.53 g
 - B) 28.4 g
 - C) 42.5 g
 - D) 66.2 g
 - E) 99.4 g

17. What is the partial pressure of gas B (in torr) if the total pressure of a mixture with the composition shown below is 825 torr?



- A) 330. torr
B) 495 torr
C) 825 torr
D) 1320 torr
E) The temperature of the system is required to answer this question.
18. An unknown gas effuses at a rate that is 1.5 times faster than chlorine gas (at the same temperature). What is the molar mass of the unknown gas?
- A) 15.76 g/mol
B) 31.51 g/mol
C) 41.27 g/mol
D) 106.35 g/mol
E) 159.53 g/mol

19. The Maxwell-Boltzmann diagram below shows the distribution of velocities for three samples of *different gases at different temperatures*. Which statement is true about the samples of gas and the graph?



- A) Curve I represents the O₂ sample.
B) Curve III represents the He sample.
C) Curve II represents the Ar sample.
D) Curve III represents the Ar sample.
E) The curves can't be correlated to the samples because both the temperature and the gases are different in each sample.
20. A given system loses a total of 776 J of internal energy while losing 325 J of heat to the surroundings. Which statement is true regarding the system?
- A) The system does 451 J of work.
B) The system has 126 J of work done on it.
C) The system does 126 J of work on the surroundings.
D) The system has 451 J of work done on it.
E) The system does 776 J of work on the surroundings.
21. Which form of the exam do you have?
- A)
B)